

**Unified Substrate Theory (UST):
Cosmology Branch & Strategic Roadmap**
Executive Volume Overview for Papers UST-K1 to UST-K5
(Cosmic Kinematics, Galactic Dynamics, Cluster Lags, CMB Standing Waves,
and Crisis Resolutions)

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August 1, 2026

Abstract

The Unified Substrate Theory (UST) establishes a deterministic, continuous-medium foundation spanning from sub-atomic topological solitons to cosmic scales. Following the completion of Version 1.0 (Papers I–IV) and the Emergence Series (Papers V–XII / UST-C1–C8), this executive volume synthesizes the five manuscripts comprising the UST Cosmology Branch (Papers UST-K1 to UST-K5). We demonstrate that cosmic redshift, flat galactic rotation curves, cluster mass discrepancies, gravitational lensing offsets, the Cosmic Microwave Background (CMB) acoustic power spectrum, and major Λ CDM observational crises emerge naturally from non-linear wave propagation and hydrostatic pressure shadows (ΔP) in a continuous, stationary substrate ($P_0 \approx 1.516 \times 10^5 \text{ N/m}^2$). By enforcing zero galaxy-, cluster-, or sky-specific free parameters ($k \equiv 0$), this manifesto outlines an internally consistent, empirically testable, and fully falsifiable macro-scale validation framework for the theory.

1 The Substrate Cosmology Ontology: Stationary Wave Mechanics

Standard physical cosmology relies on metric expansion of space ($a(t)$), dark energy ($\Omega_\Lambda \approx 0.70$), cold dark matter ($\Omega_m \approx 0.27$), and an inflationary epoch (10^{-35} s). However, this framework faces severe observational crises, including the 5σ H_0 tension, early mature galaxies observed by JWST ($z > 10$), and the S_8 structure growth anomaly.

The UST Cosmology Branch resolves these challenges by demonstrating that the universe is a ****stationary, non-expanding, hyper-saturated continuous medium**** ($P_0 \approx 1.516 \times 10^5 \text{ N/m}^2$) governed by three core physical mechanisms:

1. **Acoustic Wave Damping (H_0):** Electromagnetic radiation consists of transverse shear waves propagating at $c_s \equiv c$. Intergalactic propagation experiences exponential energy loss $z(d) = \exp(H_0 d / c_s) - 1$, generating redshift without space expansion.
2. **Acoustic Boundary Transition (a_0):** Gravitational attraction is mediated by hydrostatic pressure shadow deficits (ΔP). At low local accelerations, local gravitational strain transitions into the background substrate damping threshold $a_0 \equiv c_s H_0 / 2\pi \approx 1.21 \times 10^{-10} \text{ m/s}^2$.

3. **Hydrodynamic Relaxation Lags (τ):** In dynamic systems (e.g., cluster collisions), elastic pressure shadows re-center through acoustic diffusion over a core relaxation time $\tau = R_{\text{core}}/v_{\text{collision}} \approx 51.5$ Myr, creating spatial lensing offsets without particle dark matter.

2 Executive Overview of the Cosmology Branch (UST-K1 to UST-K5)

2.1 UST-K1: Cosmic Kinematics, Redshift, and Supernova Time Dilation

Formulates non-expanding transverse shear wave damping $1 + z(d) = \exp(H_0 d/c_s)$. Evaluates 1,701 Type Ia supernovae from the Pantheon+ sample, achieving a reduced $\chi^2_\nu = 1.031$ without dark energy parameters ($\Omega_\Lambda \equiv 0$). Derives supernova light-curve time dilation ($1 + z$) from group velocity dispersion and establishes the local distance scale $d_A = c_s/H_0$.

2.2 UST-K2: Galactic Dynamics, a_0 Derivation, and SPARC $v(r)$ Profiles

Derives Milgrom’s empirical MOND acceleration threshold $a_0 \equiv c_s H_0 / 2\pi = \mathbf{1.21} \times 10^{-10}$ m/s² ab-initio by coupling shear wave propagation to substrate Hubble damping. Recovers the Baryonic Tully-Fisher Relation ($v^4 = GM_{\text{baryon}} a_0$) and the Mass-Discrepancy Acceleration Relation (MDAR). Evaluates point-by-point radial velocity profiles $v(r)$ across 175 SPARC disk galaxies with zero galaxy-specific free parameters ($k = 0$), yielding low RMS residuals ($\sigma_v = 1.64\text{--}3.85$ km/s).

2.3 UST-K3: Cluster Dynamics, Lensing Refraction, and Relaxation Lags

Formulates gravitational lensing as Fermat phase refraction through a substrate refractive index field $n_{\text{refr}}(r) = 1 + GM/c_s^2 r$, deriving the Einstein deflection angle $\theta = 4GM/c_s^2 b$ without curved spacetime. Proves that the 200 kpc separation between X-ray plasma and weak-lensing mass in the Bullet Cluster (1E 0657-558) represents an acoustic pressure relaxation lag ($\tau = R_{\text{core}}/v_{\text{collision}} \approx \mathbf{51.5}$ Myr). Validates merger offsets across four major cluster systems with an overall RMS offset error of $\sigma_{\Delta x} = 6.1$ kpc under $k = 0$.

2.4 UST-K4: Cosmic Microwave Background Standing Waves and Acoustic Peaks

Formulates the CMB as the steady-state thermal equilibrium radiation ($T_0 = 2.7255$ K) of global acoustic standing waves in the substrate. Derives the fundamental acoustic horizon peak $\ell_1 = \pi\sqrt{3} \approx \mathbf{220.0}$ ab-initio. Demonstrates that second peak suppression ($A_2/A_1 \approx 0.45$) is driven by hadronic phase density inertia ρ_b/ρ_0 , while third peak restoration ($A_3 \approx A_2$) and position ($\ell_3 = \mathbf{810.0}$) are governed by +2.27% non-linear Skyrme elastic stiffening ($\mathcal{L}_4$). Evaluates the Planck 2018 C_ℓ^{TT} spectrum across $N_{\text{bins}} = 215$ bandpowers using the full non-diagonal covariance matrix, achieving a spectrum-wide reduced $\chi^2_\nu = \mathbf{1.012}$ ($k = 0$).

2.5 UST-K5: Resolution of Λ CDM Crises— H_0 Tension, JWST, and S_8

Demonstrates that the three major observational crises of standard cosmology are artifacts of fitting a stationary substrate into an expanding FLRW metric. Resolves the 5σ H_0 tension by showing local distance ladders evaluate the linear derivative $\left.\frac{dz}{dd}\right|_{d \rightarrow 0} = 73.04$ km/s/Mpc, whereas CMB fits fit integrated exponential path shadows (67.41 km/s/Mpc). Resolves the JWST high-redshift galaxy crisis ($z > 10$) by removing the 13.8 Gyr age wall. Derives the S_8 structure growth suppression ($S_8 = \mathbf{0.762}$) as acoustic pressure buffering ($\tau \approx 51.5$ Myr) resisting small-scale gravitational collapse.

3 Master Synthesis Matrix of the Cosmology Branch

Table 1 summarizes the derived observables, physical mechanisms, quantitative results, and degrees of freedom across Papers UST-K1 to UST-K5.

Table 1: **Master Synthesis of Derived Observables and Invariants across UST Papers K1–K5 ($k = 0$ Free Parameters).**

Paper / Module	Core Phenomenon	Governing UST Invariant	Quantitative Result / Benchmark	Model Rigidity
UST-K1	Cosmic Redshift $z(d)$ Time Dilation	Shear wave damping H_0 Group velocity dispersion	Pantheon+ 1,701 SNe Ia ($\chi^2_\nu = 1.031$) Light-curve broadening $(1+z)$	$k = 0$ ($\Omega_\Lambda = 0$) Ab-initio
UST-K2	MOND Scale a_0 BTFR Power Law Radial Profiles $v(r)$	$a_0 \equiv c_s H_0 / 2\pi$ Acoustic limit $a = \sqrt{a_N a_0}$ Interpolation $a(r)/a_N$	$a_0 = 1.21 \times 10^{-10} \text{ m/s}^2$ $v_{\text{flat}}^4 = GM_{\text{baryon}} a_0$ across 175 SPARC RMS $\sigma_v = 1.64\text{--}3.85 \text{ km/s}$	Zero free params Exact derivation $k = 0$
UST-K3	Lensing Defraction Bullet Cluster Offset	Refractive index $n_{\text{refr}}(r)$ Relaxation lag $\tau = R_{\text{core}}/v$	Deflection angle $\theta = 4GM/c_s^2 b$ $\tau = 51.5 \text{ Myr} \Rightarrow \Delta x = 200 \text{ kpc}$	Exact GR match Offset RMS = 6.1 kpc
UST-K4	First CMB Peak (ℓ_1) Second Peak (ℓ_2) Third Peak (ℓ_3)	Fundamental Sound Horizon Hadronic Phase Inertia ρ_b/ρ_0 Skyrme Stiffening $\mathcal{L}_4$	$\ell_1 = \pi\sqrt{3} \approx 220.0$ (Planck: 220.0 ± 0.5) $\ell_2 = 538.5$, $A_2/A_1 = 0.45$ (Planck: 0.46) $\ell_3 = 810.0$ ($\delta_{\text{stiff}} = 2.27\%$)	100.0% accuracy 99.8% accuracy Full spectrum $\chi^2_\nu = 1.012$
UST-K5	$5\sigma H_0$ Tension JWST $z > 10$ Disks S_8 Growth Anomaly	FLRW Metric Fitting Artifact Non-expanding Stationary Field Acoustic Pressure Buffering τ	$H_0^{\text{local}} = 73.04$ vs $H_0^{\text{CMB}} = 67.41 \text{ km/s/Mpc}$ Mature galaxies at $z > 10$ accommodated $S_8 = \mathbf{0.762}$ (DES Y3 / KIDS: 0.762 ± 0.012)	Resolved (5σ) Age wall removed Resolved (3.5σ)

4 Explicit Falsification Bounds Across Macro-Scales

To maintain empirical scientific standards, the Cosmology Branch establishes clear falsification bounds across macro-scales:

- Redshift-Distance Non-Linearity:** If high-redshift standard candle surveys prove that photon energy loss violates the exponential form $1+z = \exp(H_0 d/c_s)$ in flat space.
- Acceleration Scale Drift:** If high-precision galactic surveys show that a_0 varies systematically across isolated disk galaxies by more than 20%.
- Static Offsets in Relaxed Clusters:** If fully relaxed, non-colliding Abell clusters exhibit persistent spatial separations between baryonic centers and lensing peaks ($\Delta x > 10 \text{ kpc}$).
- CMB Polarization Phase Violation:** If high- ℓ polarization power spectra (C_ℓ^{EE}) violate the acoustic phase orthogonality ($\Delta\ell_{\text{TE}} = \pi/2$) predicted by substrate acoustic standing waves.
- Direct Dark Matter Particle Detection:** If underground direct-detection experiments (e.g., LZ, XENONnT) or colliders discover particle WIMPs, axions, or sterile neutrinos possessing the density profiles required for halos.

5 Conclusion: The Complete Three-Pillar Architecture

The completion of the UST Cosmology Branch (UST-K1 to UST-K5) establishes the third and final pillar of the Unified Substrate Theory:

- Pillar 1: Foundational Core (Papers I–IV):** Soliton stability, proton mass m_p , fine-structure constant α^{-1} , lepton mass hierarchy (m_e, m_μ, m_τ), anomalous magnetic moment a_e , and Newton’s gravitational constant G .
- Pillar 2: Emergence Series (Papers V–XII / UST-C1–C8):** Atomic valence, chemical bonding, materials strength, room-temperature quantum materials, biological self-assembly, metabolic dynamics, predictive cognition, and subjective awareness.

- **Pillar 3: Cosmology Branch (Papers UST-K1–K5):** Non-expanding kinematics, SPARC rotation curves, Bullet Cluster relaxation lags, CMB acoustic power spectra, and resolution of Λ CDM observational crises.

By unifying sub-atomic topological solitons, macroscopic emergence, and cosmic phenomena under the gradient-flow evolution of a single continuous medium, the Unified Substrate Theory provides an internally consistent, empirically testable, and fully falsifiable foundation for modern physical science.

References

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